Course milestone M4: Research Contract v0 and Permission Determination

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m04.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M04**. Brightspace carries the deadline.

This milestone presents **Book Milestone 4: Your research contract, v0** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

Declare your contract out loud at the Friday studio

That Friday studio is a contract declaration round. You present before you submit.

1. Prepare a three-minute spoken declaration of Contract v0. Walk your listeners through the fields in order.
2. Take one question from each listener. Any field is fair game, so know why you wrote every one of them.
3. Write down anything a question changed in your thinking, and fix it before you submit.
4. Submit the same day, after the studio.

Draw the model as a three-node DAG

Your contract carries a picture of the world it assumes. A DAG is a diagram where each node is a variable and each arrow claims that one variable could directly affect another.

1. Draw exactly three nodes: the thing that varies (your treatment), your outcome, and one confounder.
2. Give the confounder an arrow into both the treatment and the outcome.
3. Be ready to name the real-world mechanism behind each arrow.
4. Put the drawing in your PDF as a legible image. A photographed hand drawing is fine, and so is the studio notebook figure edited to your project.

If your question is descriptive, you still draw it. The third node is then the trait that bends both what enters your data and what you measure.

How these two are judged

The declaration is judged on craft and communication: three minutes, clear, and able to answer a question on any field. The DAG is judged with contract completeness: three nodes, a real confounder arrowed into both, and arrows you can explain aloud.

Milestone 4: Your research contract, v0

Open the milestone workbook in Colab

This chapter closes Studio 4: Declare and diagnose provisionally. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Research Contract v0: objective, target estimand, population, setting and time, data strategy, a provisional operationalization you mark for revision, warrant, answer strategy and uncertainty statement, plus the design properties you diagnosed (its bias, its wobble, and how often it would detect what you are looking for), your permission status, and a redesign record. Measurement proper is taught and assessed in Studio 6; here you only commit to a starting choice and say why it is defensible for now.

What you bring

You also carry forward the last milestone's artifact: **Milestone 3: Your evidence base**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 10: Model, Inquiry, Data Strategy, and Answer Strategy: your MIDA declaration: model, inquiry, data strategy, and answer strategy, with the alignment sentence that ties them , carrying population, setting and time, the provisional operationalization, and the warrant into Contract v0.
- Lesson 11: Uncertainty Before You Need It: your estimand and estimator, the repeat-run story behind them, your dependence structure, and your uncertainty sentence next to its wrong twin.
- Lesson 12: Declaring and Diagnosing a Research Design: your design diagnosis: bias, wobble, and power read from simulation, one redesign, and the honest call that followed.
- Lesson 13: Research Ethics and Data Governance: your declared permission status, your minimised column list, your AI boundary, and your four governance decisions.

The practice

1. Write the four MIDA parts of your design so a stranger could run them. (MIDA is Blair, Cooper, Coppock, and Humphreys' framework, developed at book length in RDSS.)
2. State your estimand and estimator, and describe in words what would differ if the procedure ran again.
3. Diagnose the design: what is its bias, how much does it wobble, and how often would it detect what you are looking for.
4. Run the permission determination and declare one status; if it is anything but cleared, name the authority and the date you asked.
5. Minimise your columns against the declared analysis, and check what a re-identification test would find.
6. Write the redesign record - what you changed after diagnosing, and why. A diagnosis that changed nothing is a diagnosis you should distrust.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** The permission status produced here gates every later studio; nothing downstream may proceed past a stop.
- **Evidence, provenance, and reproducibility.** Your data strategy names actual sources from the Studio 3 registry, not source types.
- **AI activity, verification, and human decisions.** Diagnosis is delegable; the choice of what to fix is not.
- **Uncertainty, claim boundary, and revision history.** This is where your uncertainty statement is first written down and first tested.

Where this milestone sits

You arrived carrying Milestone 3: Your evidence base. Milestone 5 commits the contract to one research pathway and writes what that pathway can and cannot establish. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Route diagnosis, measurement changes, permission rulings, and the realized data each produce a new Contract version with its reason.

How this milestone is assessed

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Research Contract v0: objective, target estimand, population, setting and time, data strategy, a provisional operationalization marked for revision, warrant, answer strategy and uncertainty statement, plus the design properties you diagnosed (its bias, its wobble, and how often it would detect what you are looking for), your permission status, and a redesign record	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

> ! Blocking gate
>
> No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.